

1. **Experiment:** Selective Inhibition of IL-6 Trans-signaling and Preservation of Classical Signaling by STAR-Trap Macrophages
2. **Time:** 2026.08.12-2026.08.31
3. **Member:** Qichang Liu, Kaili Jin, Zhangdong Liu, Yuxuan Wang, Xinru Yuan
4. **Materials:** IL-6, sIL-6R α , IL-6/sIL-6R α complex, WT-M ϕ , STAR-Trap macrophages (THP-1-derived), HUVEC cells (for trans-signaling assay), U266 cells (for classical signaling assay), Tocilizumab, complete medium (RPMI-1640 with 10% FBS and 1% penicillin-streptomycin), FACS buffer (PBS with 1% FBS), fixation/permeabilization kit, antibodies against pSTAT3, MCP-1, and ICAM-1.
5. **Method:**

(1) Inhibition of IL-6 trans-signaling:

- ① The following experimental groups were set up:

Group	Description
Group 1	complex
Group 2	complex + WT-M ϕ
Group 3	complex + STAR-Trap

- ② Different macrophages were incubated with IL-6/sIL-6R α complex.
 ③ Cell-free supernatants were collected.
 ④ The supernatants were added to the HUVEC cell culture system.
 ⑤ IL-6 trans-signaling in endothelial cells was detected by measuring pSTAT3, MCP-1, and ICAM-1.

(2) Preservation of IL-6 classical signaling:

- ① U266 cells were used for the classical signaling assay. The expression of mIL-6R α and gp130 on U266 cells was confirmed by flow cytometry prior to the experiment.

- ② The following experimental groups were set up:

Group	Description
Group 1	IL-6
Group 2	IL-6 + WT-M ϕ
Group 3	IL-6 + STAR-Trap
Group 4	IL-6 + Tocilizumab (positive control for classical signaling inhibition)

- ③ Different macrophages were incubated with IL-6.
 ④ Cell-free supernatants were collected.
 ⑤ The supernatants were added to the U266 cell culture system.
 ⑥ IL-6 classical signaling in U266 cells was detected by measuring pSTAT3, MCP-1, and ICAM-1.

(3) Flow cytometric analysis:

- ① Cells were harvested and washed twice with FACS buffer.
 ② For pSTAT3 detection, cells were fixed and permeabilized (including nuclear permeabilization) using a Foxp3/Transcription Factor Staining Buffer Set or equivalent, followed by intracellular staining with anti-pSTAT3 antibody for 30 min at 4 °C in the dark.
 ③ For ICAM-1 detection, cells were stained with the corresponding antibody on the cell surface according to the standard staining protocol.
 ④ For MCP-1 detection, the culture supernatant was collected and measured by ELISA or CBA (no permeabilization required).
 ⑤ Flow cytometry was performed, and the mean fluorescence intensity (MFI) and positive cell percentages were recorded for each group.

6. Result:

- (1) Inhibition of IL-6 trans-signaling: The expression levels of pSTAT3, MCP-1, and ICAM-1 in endothelial cells treated with supernatants from different groups are summarized below.

Group	pSTAT3	MCP-1	ICAM-1
complex	↑↑↑	↑↑↑	↑↑↑
complex + WT-Mφ	↑↑↑	↑↑↑	↑↑↑
complex + STAR-Trap	↓↓↓	↓↓↓	↓↓↓

These results indicate that STAR-Trap macrophages selectively inhibit IL-6 trans-signaling.

(2) Preservation of IL-6 classical signaling: The expression levels of pSTAT3, MCP-1, and ICAM-1 in U266 cells treated with supernatants from different groups are summarized.

Group	pSTAT3	MCP-1	ICAM-1
IL-6	↑↑↑	↑↑↑	↑↑↑
IL-6 + WT-Mφ	↑↑↑	↑↑↑	↑↑↑
IL-6 + STAR-Trap	↑↑↑	↑↑↑	↑↑↑
IL-6 + Tocilizumab	↓↓↓	↓↓↓	↓↓↓

These results indicate that STAR-Trap macrophages preserve IL-6 classical signaling, as opposed to the pan-inhibition observed with Tocilizumab.

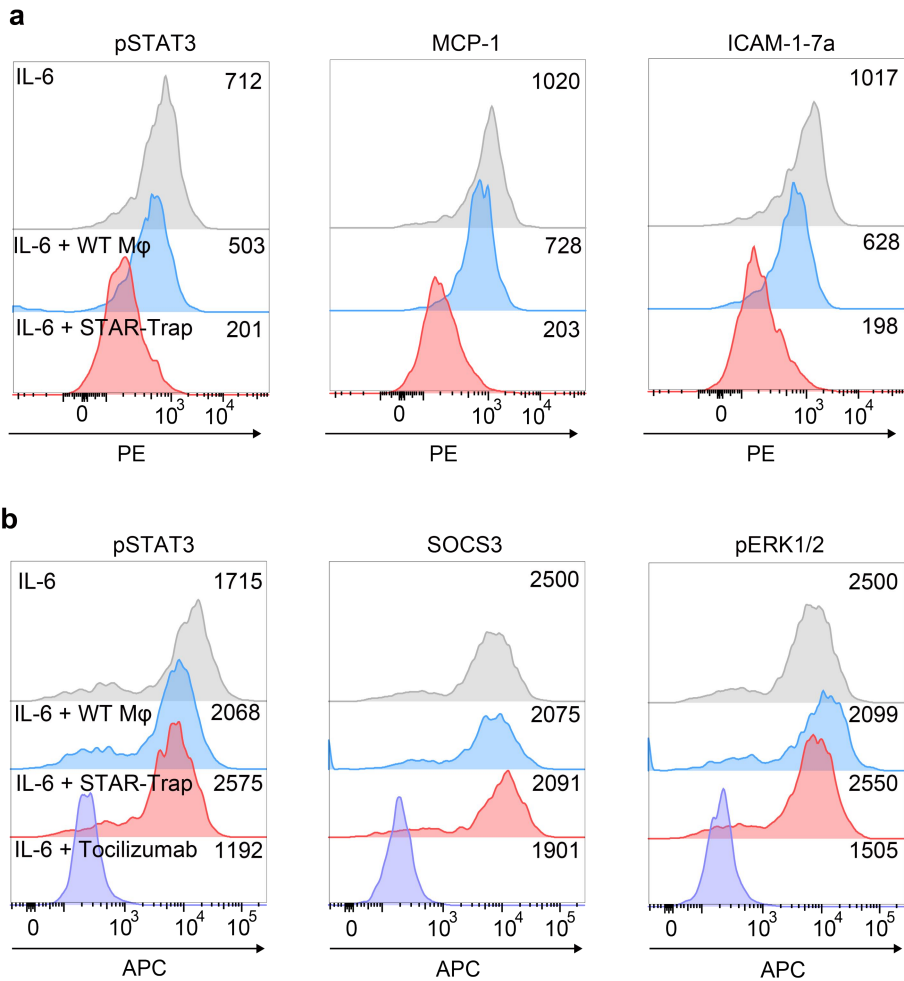


Figure 1. STAR-Trap attenuates IL-6 trans-signaling while preserving classical signaling. **a**, Representative flow-cytometry histograms showing IL-6 trans-signaling responses. The IL-6/sIL-6Rα complex was incubated alone or preincubated with WT or STAR-Trap. After removal of the macrophages, the resulting cell-free supernatants were applied to gp130-positive responder cells with low or undetectable mIL-6Rα expression. Intracellular pSTAT3 and the downstream inflammatory markers MCP-1 and ICAM-1 were subsequently measured. Grey, IL-6/sIL-6Rα complex alone; blue, complex preincubated with WT macrophages; red, complex preincubated with STAR-Trap. **b**, Representative flow-cytometry histograms showing classical IL-6 signaling responses. Free IL-6 was incubated alone or preincubated with WT or STAR-Trap, and the resulting cell-free supernatants were applied to responder cells expressing both mIL-6Rα and gp130. Intracellular pSTAT3, SOCS3 and pERK1/2 were measured, with Tocilizumab included as an IL-6R-blocking control. Grey, IL-6 alone; blue, IL-6 preincubated with

WT macrophages; red, IL-6 preincubated with STAR-Trap; purple, Tocilizumab-treated control. Numbers indicate the median fluorescence intensity of the corresponding cell populations.